

ConstructVision AI: An Evidence-First Digital-Twin Architecture for AI-Assisted Civil Infrastructure Inspection

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Abstract

Civil infrastructure inspection remains fragmented across photographs, paper observations, spreadsheets, sensor streams and engineer judgement. ConstructVision AI is an evidence-first digital-twin prototype that explores a unified command centre for structural context, inspection triage, telemetry, preliminary mechanics and repair costing. The implemented system uses Streamlit for the interaction layer, OpenCV for rule-based image analysis, Three.js and Plotly for parametric three-dimensional visualisation, Pydantic contracts for typed asset and telemetry records, and a worker-readiness boundary for future reconstruction, finite-element, computational-fluid-dynamics and flood jobs. The prototype converts image contours into approximate crack dimensions through an explicit pixel-to-millimetre calibration, classifies severity with transparent thresholds, and computes screening indicators for gravity, flexure, wind pressure and hydrostatic pressure. It also provides cost estimation from area, material coverage, urgency, labour, scaffolding and GST. The contribution is therefore architectural and methodological: it demonstrates how visual outputs, assumptions, uncertainty and human review can be kept connected rather than allowing a compelling 3D view to be mistaken for certified engineering evidence. YOLOv8-based detection, ESG carbon forecasting and four-dimensional seismic response are specified as validated extension paths, not represented as capabilities already measured in the current checkout. The paper proposes a mathematical integration layer for these extensions, including detection loss, seismic response, energy conversion and embodied-carbon accounting. A rigorous evaluation protocol is defined around precision, recall, mAP, calibration error, latency, reconstruction error, forecast error and engineer agreement. The resulting system is a responsible foundation for future deployment in civil infrastructure inspection.

Keywords: civil infrastructure inspection; digital twin; computer vision; structural health monitoring; Streamlit; seismic simulation; embodied carbon; evidence provenance.

1. Introduction

Inspection teams must combine visual observations, component identity, location, condition, environmental context and recommended action. In practice, these facts are often stored in different artefacts. A defect can be photographed on site, described in a log, measured later and finally transferred to a report without a stable link to the building element or the model revision used for interpretation. Manual inspection remains essential, but it is time-intensive, difficult to standardise and vulnerable to inconsistent documentation.

ConstructVision AI addresses this workflow problem through a browser-based command centre. Its current implementation is a prototype, not a safety-certification system. The project links a parameterised structural frame, inspection images, simulated telemetry, preliminary mechanics, cost estimation and future analysis jobs. Its central design principle is evidence before spectacle: a parametric model is not a calibrated reconstruction; a visual animation is not a validated solver result; and an AI flag is not a professional sign-off.

1.1 Research gap

Prior research has separately advanced computer-vision defect detection, structural-health monitoring, building information modelling and digital twins. The unresolved engineering gap is operational integration under uncertainty. A useful inspection twin must preserve source imagery, calibration, component identity, model revision, sensor quality, analysis status and reviewer responsibility in one auditable chain. The present work contributes a reference architecture and transparent screening mathematics for this integration.

1.2 Contributions

1. An evidence-first architecture connecting capture, inspection, spatial context, telemetry, screening and reporting.
2. A traceable implementation analysis grounded in the current Python and Streamlit codebase.
3. Mathematical formulations for the implemented screening calculations and proposed YOLOv8, seismic and carbon extensions.
4. An evaluation protocol that separates measured performance from simulated demonstration values.
5. A governance boundary that prevents unavailable external solvers from being represented as completed engineering analyses.

2. Literature review

Computer-vision crack detection research has progressed from thresholding and edge operators to supervised detection, segmentation and multimodal assessment. Reviews identify acquisition quality, lighting, material texture, scale and dataset bias as persistent limitations [1]-[4]. One-stage detectors are attractive for field triage because they provide low-latency predictions, but a deployment requires a labelled dataset, train/validation/test partitioning, class definitions, calibration strategy and false-negative analysis.

Digital twins in construction are more than three-dimensional models. They combine a physical asset, a digital representation, data exchange and a lifecycle process. Reviews emphasise interoperability, synchronisation, data quality, semantics, cybersecurity and the difficulty of maintaining a reliable relationship between model and asset [5]-[8]. Structural-health monitoring adds time-series sensing and damage-sensitive features, but sensor drift, environmental variability and missing data must be handled explicitly [9].

Seismic response analysis is normally conducted with validated structural models, ground-motion records and code-defined combinations. A browser animation can communicate a scenario but cannot replace modal analysis or nonlinear time-history analysis. Similarly, carbon accounting requires defined system boundaries, activity data, emission factors and reporting periods; a dashboard label alone is not an ESG inventory [10]-[12].

The research gap is therefore not the absence of individual algorithms. It is the absence of a transparent bridge from an observation to a reviewable engineering action. ConstructVision's current prototype makes that bridge explicit, while leaving high-stakes algorithms behind versioned validation gates.

3. Methodology

3.1 Current inspection pipeline

`pages/6\_■\_AI\_Inspection.py` accepts a user image or synthesises a sample crack image. The current pipeline is:

1. Convert RGB pixels to grayscale.
2. Apply Gaussian blur with a 5×5 kernel.
3. Apply Canny edge detection with thresholds 50 and 150.
4. Extract external contours.
5. Remove contours with area $A_c \leq 20$ pixels.
6. Compute a bounding rectangle (x, y, w, h) .
7. Convert dimensions using a user-supplied scale s in mm/px:

$$L_{\text{mm}} = s \max(w, h), \quad W_{\text{mm}} = s \min(w, h).$$

8. Assign severity using the implemented rule:

$$C(W_{\text{mm}}) =$$

$\begin{cases}$

$\text{CRITICAL}, & W_{\text{mm}} > 2.0 \backslash$

$\text{\text{WARNING}}, & 1.0 < W_{\text{mm}} \leq 2.0 \backslash$

$\text{\text{SAFE}}, & W_{\text{mm}} \leq 1.0.$

$\text{\text{end{cases}}}$

The page displays a confidence-like value $\min(0.95, 0.60 + A_c/1000)$, but this is a heuristic display value, not a calibrated probability. It must not be reported as model confidence in an empirical paper.

3.2 Proposed YOLOv8 extension

The repository contains a settings label for YOLOv8-X/N and a `models/yolov11.pt`` file, but no YOLO inference call, dataset loader or trained YOLOv8 checkpoint is wired into the active inspection page. The proposed production path is an anchor-free, decoupled-head detector with a backbone-neck feature pyramid and defect classes such as crack, spalling, exposed reinforcement, honeycombing, leakage and corrosion.

For a predicted box $\hat{b}=(\hat{x}, \hat{y}, \hat{w}, \hat{h})$ and ground-truth box b , a composite objective can be written as:

$$\mathcal{L} = \lambda_{\text{box}} \mathcal{L}_{\text{CIoU}} + \lambda_{\text{cls}} \mathcal{L}_{\text{BCE}} + \lambda_{\text{dfl}} \mathcal{L}_{\text{DFL}},$$

where

$$\mathcal{L}_{\text{CIoU}} = 1 - \text{IoU} + \frac{\rho^2(\hat{c}, c)}{d^2} + \alpha v,$$

and v measures aspect-ratio inconsistency. For class imbalance, focal weighting may be added:

$$\mathcal{L}_{\text{focal}} = -\alpha_t (1 - p_t)^\gamma \log(p_t).$$

These equations describe a proposed training specification; they are not evidence that this checkout trained YOLOv8.

3.3 Digital-twin geometry and telemetry

`modules/digital_twin/engine.py`` defines `TwinInputs`` containing floors, bays, bay spacing, storey height, concrete strength, member dimensions, wind speed, rainfall, flood depth and live load. Frame nodes are generated by:

$$\mathbf{n}_{\text{xyz}} = (x, b, y, b, z, h_s),$$

for integer grid indices. Members connect vertical nodes as columns and adjacent same-level nodes as beams. The deterministic telemetry generator samples temperature, wind, strain and tilt. With a random generator seed, it computes:

$$T_t = 25 + 6 \sin(\theta_t) + \epsilon_T,$$

$$V_t = \max\{0, 8 + 5 \sin(\phi_t) + \epsilon_V\},$$

$$\epsilon_t = 165 + 1.2 T_t + 2.3 V_t + \epsilon_{\text{varepsilon}},$$

$$\epsilon_{\theta_t} = 0.15 + 0.012 V_t + \epsilon_{\text{vartheta}}.$$

These values are synthetic telemetry for interface demonstration.

3.4 Preliminary structural screening

For an interior lower-storey column, tributary area, gravity action, column area and axial stress are:

$$A_t = b^2,$$

$$P = (6.0 + q_L) A_{n_f},$$

$$A_c = (d_c/1000)^2,$$

$$\sigma_a = P / (1000 A_c).$$

The code uses $0.40 f_c$ as a screening capacity indicator. The simplified beam line load and moment are:

$$w = (6.0 + q_L) b, \quad M = \frac{w b^2}{8},$$

with section-modulus indicator $Z=(d_b/1000)^{3/6}$ and $\sigma_f=M/(1000Z)$. Wind and hydrostatic pressures are:

$$q_w=0.613V^2/1000\text{ kPa}, \quad p_h=9.81h_f\text{ kPa}.$$

The utilisation index is:

$$U=\max\left(\frac{\sigma_a}{0.40f_c}, \frac{\sigma_f}{0.60f_c}\right).$$

The code returns `Review required` when $U \geq 0.8$. These are transparent screening indicators, not code-compliant design checks.

3.5 Proposed 4D seismic simulation

The current UI includes a `Seismic Oscillation` selector, but the checkout has no seismic ground-motion record or dynamic solver. A future validated module can define a base-excited multi-degree-of-freedom model:

$$M\ddot{u}+C\dot{u}+Ku=-M\ddot{u}_g(t),$$

where u is relative displacement and $\ddot{u}_g(t)$ is ground acceleration. A modal approximation is:

$$u(z,t)=\sum_{r=1}^m \phi_r(z) q_r(t),$$

with modal coordinates satisfying:

$$\ddot{q}_r+2\zeta_r\omega_r\dot{q}_r+\omega_r^2q_r=-\Gamma_r\ddot{u}_g(t).$$

For a visualised inter-storey drift:

$$\Delta_i(t)=u_i(t)-u_{i-1}(t), \quad \theta_i(t)=\frac{\Delta_i(t)}{h_i}.$$

The simulation should use recorded or code-compatible motions, validated mass/stiffness matrices, time-step convergence and engineer review. It must not be presented as implemented merely because the UI animates sway.

3.6 Proposed ESG carbon forecasting

No carbon calculation is present in the source. A future activity-based inventory can compute:

$$E_{CO_2e}=\sum_{j=1}^N A_j EF_j CF_j,$$

where A_j is activity data, EF_j an emission factor and CF_j a conversion or global-warming-potential factor. A time-series forecast can model monthly emissions as:

$$\hat{E}_{t+h}=f(E_{1:t}, x_{1:t+h}; \Theta),$$

with uncertainty intervals $[\hat{E}_{t+h}^L, \hat{E}_{t+h}^U]$. The system should retain source, factor version, boundary, unit and reporting period for every term.

4. Implementation and system architecture

The platform is a Streamlit multi-page experience with the following 13 functional modules:

1. Home/command centre and project portfolio.
2. Tutorial and training workflow.
3. Structural digital twin and 3D building viewer.
4. Materials knowledge library.
5. Virtual practical laboratory.
6. AI inspection and crack triage.
7. Damage analysis and repair protocol.
8. Cost estimation and bill generation.
9. Report generation and audit history.

10. IoT/sensor-node visualisation.
11. Historical inspection records.
12. Settings, calibration and governance.
13. Platform-core contracts and future worker orchestration.

The newer root ``app.py`` also exposes Portfolio, Command center, Twin studio, Capture & reconstruction, Asset health and Integration views. ``platform_core/contracts.py`` defines typed asset, capture, telemetry and analysis-job models. ``workers/dispatcher.py`` checks whether COLMAP, OpenSees and OpenFOAM executables are installed and returns ``READY`` or ``BLOCKED``; this is a valuable safety property because unavailable analyses are not fabricated.

5. Results and discussion

5.1 Verified implementation results

The code supports a runnable interface, deterministic demo telemetry, parametric frame geometry, transparent screening formulas, image contour annotation, repair-cost calculation, report-history views and typed contracts. The source also states that reconstruction, FEA, CFD and production IoT are future integrations.

5.2 Simulated demonstration metrics

The following values are planning targets for a future evaluation harness, not measured results from this checkout: detector $mAP@0.5 \geq 0.85$; defect recall ≥ 0.90 ; calibrated confidence error ≤ 0.05 ; single-image inference latency ≤ 200 ms on an edge GPU; telemetry-to-alert latency ≤ 2 s; crack-width mean absolute error ≤ 0.5 mm after calibrated capture; and carbon forecast MAPE $\leq 15\%$. They must be replaced with dataset-backed measurements before journal submission.

5.3 Discussion

The most defensible current contribution is traceability. The prototype makes assumptions visible, uses explicit formulas, exposes the calibration ratio, and keeps solver unavailability visible. Its principal weakness is that the strongest requested claims - YOLOv8 accuracy, seismic response fidelity and ESG forecasting quality - cannot yet be supported by executable evidence. The research programme should therefore prioritise labelled data, calibration, versioned model artefacts, validation cases, uncertainty reporting and professional review before adding visual complexity.

6. Conclusion and future scope

ConstructVision AI demonstrates how an infrastructure-inspection interface can connect images, geometry, telemetry, mechanics, costing and workflow governance. Its evidence-first architecture is suitable as a foundation for a validated digital twin, but the present implementation is a prototype and must not issue safety certificates or design approvals. Future work should integrate a trained detector, calibrated photogrammetry or BIM/IFC input, time-synchronised sensor streams, validated OpenSees/FEA workflows, seismic response histories, activity-based carbon accounting, role-based access control, immutable evidence storage and independent civil-engineering review.

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